

1 Results

1.1 Gathered data

- Number of data available for angiosperm and gymnosperm originally from Silvics book, with masting data, and the structure of data
 - Angiosperm (In the book: 124; with masting data: 98; Masting/Non-masting: (51/47); Gymnosperm (In the book: 64; with masting data: 61; Masting/Non-masting: (57/7))
 - Strong masting more common for gymnosperm
- Number of traits we have for different hypotheses for angiosperm and gymnosperm
 - For angiosperm, we have data for:
 - * Predation satiation: seed weight, fruit size, oil content, seed dispersal mode, seed dormancy
 - * Pollination coupling: pollination mode, reproductive type
 - * Resource matching: drought tolerance, leaf longevity
 - For gymnosperm, we only have oil content for masting species and all the gymnosperm species are wind pollinated.

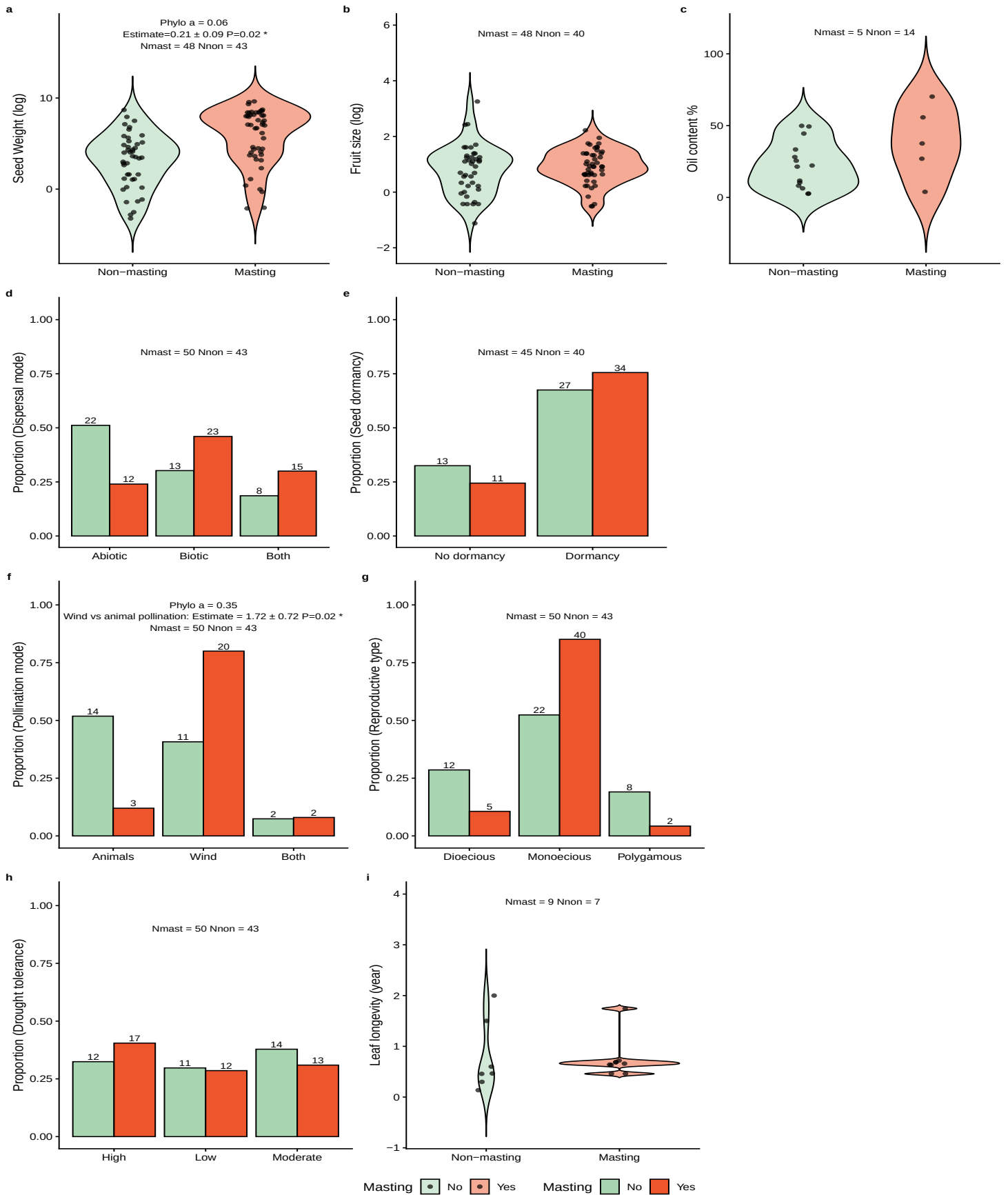


Figure 1: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf longevity and masting based on phylogenetic logistic regression for angiosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phyloglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter α , and associated P-value. (a)-(e) are traits related to predation satiation hypothesis; (f), (g) are traits related to pollination coupling hypothesis; (h), (i) are traits related to resource matching hypothesis. Legends are shared across panels.

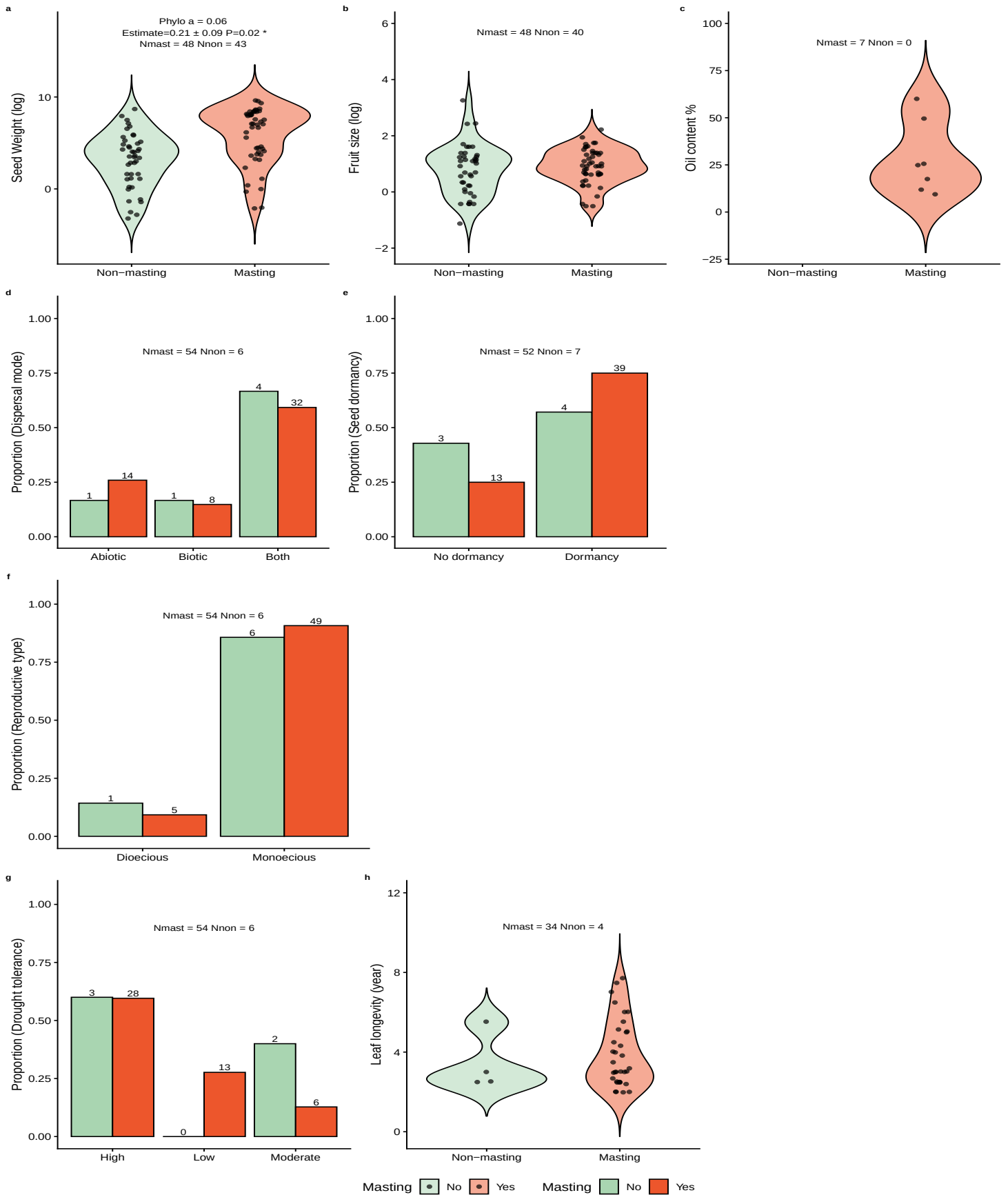


Figure 2: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf longevity and masting based on phylogenetic logistic regression for gymnosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phylglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter α , and associated P-value. (a)-(e) are traits related to predation satiation hypothesis; (f), (g) are traits related to pollination coupling hypothesis; (h), (i) are traits related to resource matching hypothesis. Legends are shared across panels.

Trait	Predictor	Estimate	SE	P	Phylo α	N
Seed dispersal	Biotic vs abiotic	-0.83	0.81	0.30	0.01	60
Seed dispersal	Both vs abiotic	-1.11	0.67	0.10	0.01	60
Seed dormancy	Dormant vs non-dormant	0.71	0.70	0.32	0.41	59
Reproductive type	Monoecious vs dioecious	-1.14	1.02	0.26	0.01	61
Seed weight (log)	logSeedWeight	-0.03	0.25	0.91	0.43	59
Fruit size (log)	logFruit	0.03	0.36	0.94	0.43	60
Seed size (log)	logSeedSize	-0.24	0.92	0.80	0.41	51
Leaf longevity (years)	Leaf longevity (years)	0.15	0.34	0.67	0.43	38
Drought tolerance	Low vs high drought tolerance	0.91	1.64	0.58	0.43	52
Drought tolerance	Moderate vs high drought tolerance	-1.56	1.06	0.14	0.43	52

Table 1: Phylogenetic Generalized Linear Model Results for Gymnosperm

Trait	Predictor	Estimate	SE	P	Phylo α	N
Dispersal mode	Biotic vs abiotic	0.55	0.53	0.30	0.06	93
Dispersal mode	Both vs abiotic	0.55	0.57	0.33	0.06	93
Pollination mode	Wind vs animal pollination	1.72	0.72	0.02	0.35	52
Pollination mode	Wind vs animal pollination and animals	1.23	1.15	0.29	0.35	52
Seed dormancy	Dormant vs non-dormant	0.03	0.50	0.95	0.10	85
Reproductive type	Monoecious vs dioecious	0.90	0.68	0.18	0.11	89
Reproductive type	Monoecious vs polygamous	-0.13	0.96	0.89	0.11	89
Seed weight (log)	logSeedWeight	0.21	0.09	0.02	0.06	91
Fruit size (log)	logFruit	0.08	0.30	0.80	0.17	88
Seed size (log)	logSeedSize	0.24	0.30	0.43	0.08	69
Oil content (%)	Oil content (%)	0.04	0.03	0.20	0.14	19
Leaf longevity (years)	Leaf longevity (years)	0.26	0.95	0.78	0.01	16
Drought tolerance	Low vs high drought tolerance	0.04	0.48	0.92	0.11	79
Drought tolerance	Moderate vs high drought tolerance	-0.13	0.46	0.79	0.11	79

Table 2: Phylogenetic Generalized Linear Model Results for Angiosperm

1.2 Results for each hypothesis

- Predation satiation: seed weight only with a small effect and only for angiosperm
 - We found only seed weight is related to masting with having bigger seeds will increase the chance of being a strong masting species Figure (1), At the mean log seed weight (4.54), the predicted probability of masting was 0.43. One unit increase in log seed weight increasing the odds of masting by approximately 22%. (??). Only within angiosperm we found this significant effect of seed weight.
- Pollination satiation: wind pollination for angiosperm
 - we found wind pollination is related to masting for angiosperm species, wind pollinated species have a significantly higher probability (59%) of being strong masting species. Gymnosperm are all wind pollinated, and gymnosperm has a greater portion of masting species (57/61).
- Resource matching: nothing significant

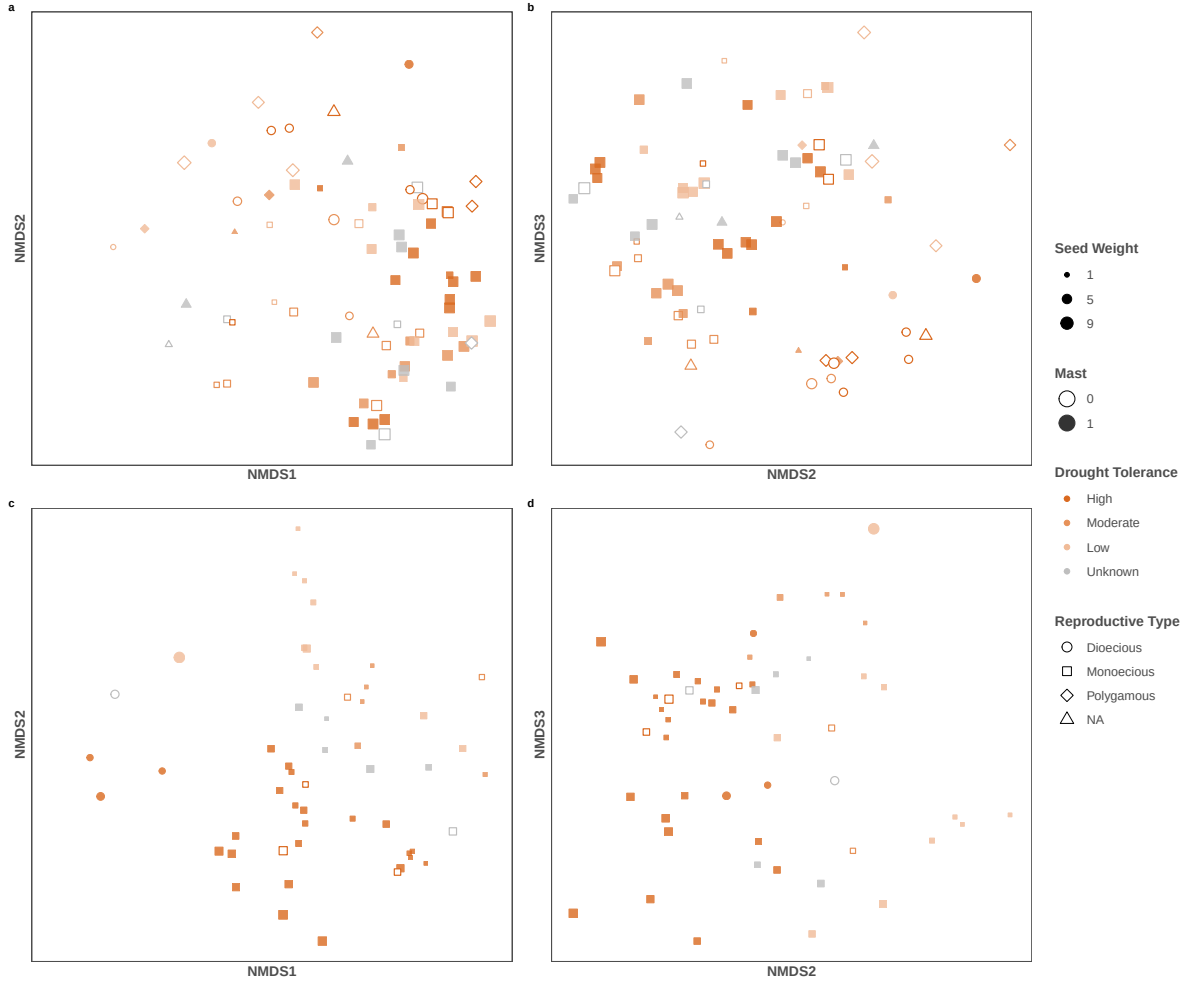


Figure 3: Non-metric multidimensional scaling (NMDS) of trait composition for angiosperm and conifer species. Each point represents a species, with position determined by NMDS axes. Points are colored by drought tolerance, filled by masting or not, shaped by reproductive type, sized by seed weight. Panels are arranged as (a) angiosperm NMDS1–NMDS2, (b) angiosperm NMDS2–NMDS3, (c) conifer NMDS1–NMDS2, and (d) conifer NMDS2–NMDS3.

1.3 Multidimension-NMDS

- No single trait drives most of the variation
 - We don't see masting aligned well along any axis in the NMDS analysis.
 - There is no strong overall trait structure for all species

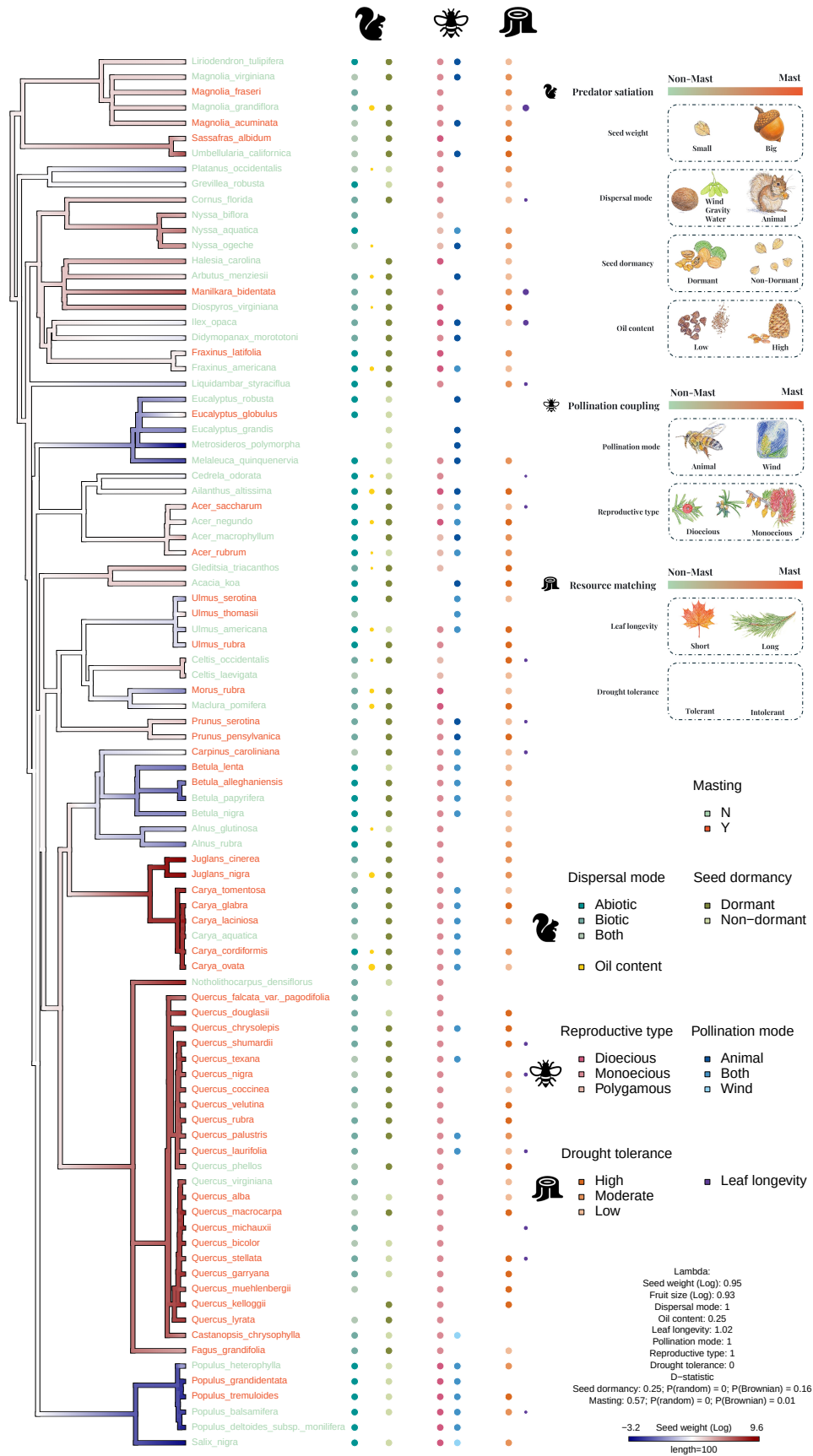


Figure 4: Phylogenetic tree for all angiosperm species included in the analysis. The color scheme of the branch is the trait value for seed weight (log). The color tip is colored by masting or not. The other traits are labeled after the tip labels with different color and size. We also show the predictions of relationship between traits and masting. We calculated the Pagel's λ for continuous traits as well as multilevel categorical traits, and calculated D-statistic for binary traits

Group	Trait	(λ)
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.02
Angiosperm	Dispersal mode	1.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	1.00
Angiosperm	Drought tolerance	0.00
Gymnosperm	Seed weight (Log)	0.97
Gymnosperm	Fruit size (Log)	0.90
Gymnosperm	Oil content	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	1.00
Gymnosperm	Drought tolerance	1.00

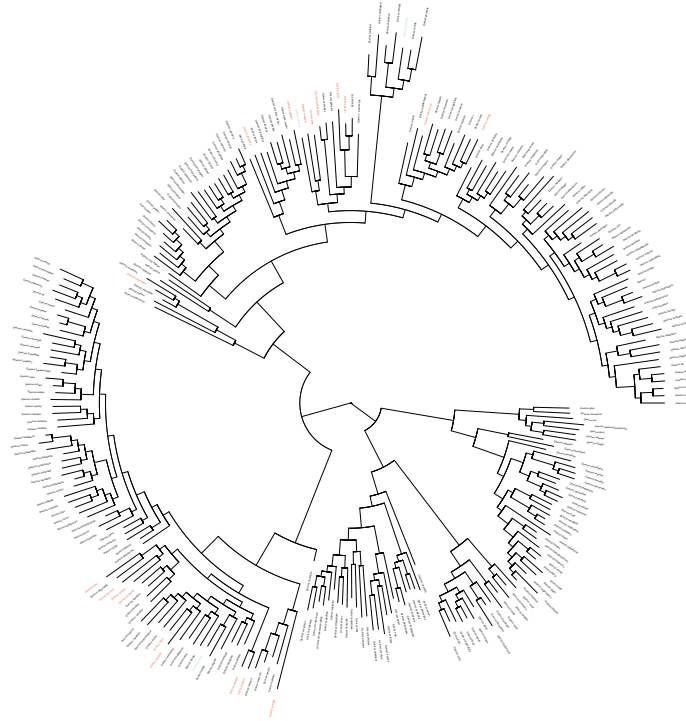
Table 3: Phylogenetic Signal (Pagel’s λ) for continous traits and multi-level categorical traits

Group	Trait	Estimated D	P(random)	P(Brownian)
Angiosperm	Seed dormancy	0.25	0	0.16
Angiosperm	Masting	0.58	0	0.01
Gymnosperm	Masting	0.51	0.02	0.10
Gymnosperm	Seed dormancy	0.98	0.41	0
Gymnosperm	Reproductive type	-0.52	0	0.89

Table 4: Phylogenetic Signal (D-statistic) for the phylogenetic signal of binary traits, with probabilities calculated under a random phylogenetic structure and under a Brownian motion phylogenetic structure. $D = 1$ indicates the trait is randomly distributed across the phylogeny, $D = 0$ indicates the trait follows a Brownian motion model, $D < 0$ indicates the trait is more clumped than expected under Brownian motion, and $D > 1$ indicates overdispersion. P random is the p-value for the null hypothesis that the trait is randomly distributed across the phylogeny (no phylogenetic signal), and P Brownian is the p-value for the null hypothesis that the trait follows a Brownian motion model (strong phylogenetic signal). A low p-value indicates rejection of the corresponding null hypothesis.

1.4 Phylogenetic patterns

- Pagel's λ for continuous traits and multilevel categorical traits in Table 3, D-statistic for binary traits in Table 4.
 - Masting shows a stronger phylogenetic signal in gymnosperms (estimated $D = 0.51$) than in angiosperms (estimated $D = 0.58$). For angiosperms, the P-value under the Brownian motion model is 0.01, indicating that masting does not significantly follow a Brownian motion model.
 - Seed weight has strong phylogenetic signal for both angiosperm ($\lambda = 0.95$) and gymnosperm ($\lambda = 0.97$). We can see the similarity in certain group in Figure 4 it's relatively reliable estimate
 - Be cautious that λ might not be comparable among traits, not only because the way λ is calculated for continuous traits and categorical traits are different, but also because λ might depend on the data structure when sampling is uneven, take leaf longevity as an example in Figure 4.
- When we know the phylogenetic results, we need to be cautious interpreting relationships between traits, it is hard to tease out traits from masting, which limits the inference)



• Masting
• Non-masting

Figure 5: Phylogenetic tree for all *Quercus* species, *Quercus* species included in this study is color coded.

1.5 Data sparsity

- Among all the *Quercus* species with well-resolved phylogeny (268), our dataset includes only 21 species, and only two of them are non-masting species. 5.
- Data is strongly nested in phylogeny, and data is sparse 4. We have to assume the sampling is even.

2 Methods

2.1 Data collection

- Data scraping
 - Data sources: Silvics of North America, Kew Seed Dataset, TRY Dataset
 - Traits selection
 - Key words used for data scraping
 - * Definition of strong masting or not
 - * Criteria for other traits
- Data cleaning
 - Cleaning of categorical traits
 - Decision for numeric traits with a range

2.2 Statistical analysis

- Main analysis for masting pattern and single trait: phyloglm
- Masting pattern with a multivariate analysis: NMDS
- Phylogenetic signal check for traits: Pagel's λ and D-statistics